Yen-Pu (Paul) Wang

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EDUCATION

University of California, San Diego (UCSD) — La Jolla, CA

Master of Science in Computer Science & Engineering (System Track); GPA: 3.77/4.0 — *Sept. 2024 - Dec. 2025*

- **Coursework:** Operating Systems, Machine Learning Systems, Distributed Computing & Systems, Virtualization and Cloud Computing, Parallel Computation, Networked Systems, Database Systems, Convex Optimization, Recommender Sys & Web Mining

National Taiwan University (NTU) — Taipei, Taiwan

Bachelor of Arts in Economics; GPA: 3.9/4.0 — *Sept. 2020 - June 2023*

- **Coursework:** Programming Design, Data Structure and Algorithms, Algorithm Design and Analysis, Operating Systems, Distributed Systems and Cloud Application Development, Machine Learning, Deep Learning, Text Mining

SKILLS

Programming Languages: Rust, Go, Python, C/C++, JavaScript, SQL, Bash

Cloud & DevOps: AWS, Azure, GCP, Kubernetes, Docker, Terraform, Flyte, Helm, gRPC, Protobuf, CI/CD, GitHub Actions, Harbor, Git

ML/AI: PyTorch, TensorFlow, LLM, Triton, CUDA, Hugging Face, Scikit-learn

Data & Storage: Kafka, Spark, PostgreSQL, Redis

PROFESSIONAL EXPERIENCE

Yahoo — Mountain View, CA

Production Engineer Intern (Systems and Infrastructure) | GCP, Vertex AI, MLOps, Docker, Python, Bash, CI/CD — *June 2025 – Sept. 2025*

- Designed and built a scalable framework of reusable templates for machine learning workflows, reducing new project setup time for engineering teams from 6 weeks to 1 week.
- Slashed LLM fine-tuning cycle time by 30% by engineering a multi-stage training pipeline template triggered through git commits, which builds cached containers and dynamically provisions ML training pipelines on Vertex AI.
- Unblocked the Mail Intelligence team by architecting a pre-merge validation pipeline that benchmarks candidate models against production baselines; automated side-by-side inference on Vertex AI prevents regressions and enables safe continuous delivery.
- Pioneered the adoption of a new internal ML Infra tool, driving its roadmap by implementing asynchronous job submission to unblock long-running ML workflows, revamping core documentation, and providing foundational feedback as its first users.

Trend Micro - #1 market share worldwide cloud-native application protection platform — Taipei, Taiwan

Software Engineer Intern (Infrastructure) | Azure, Kubernetes, Docker, Kafka, Helm, Python, Bash, CI/CD — *July 2024 – Aug. 2024*

- Re-architected a legacy system by migrating VM-based scripts to event-driven backend microservices on Kubernetes (AKS), significantly improving system reliability and auto-scaling.
- Dropped cloud compute costs by 95% and boosted ingestion efficiency by 80% by building a high-throughput Kafka data ingestion service for real-time log processing.
- Implemented a secure backend service with RBAC and query filtering for Microsoft Graph API (AAD) data retrieval, enabling secure, self-service authentication data access for 10+ engineers.
- Engineered an end-to-end CI/CD automation pipeline for a monorepo, optimizing Docker builds, vulnerability scanning (Harbor), and ACR deployments to reduce release time.

Shopee - leading e-commerce platform in Southeast Asia — Taipei, Taiwan

Data Engineer Intern | Python, PostgreSQL, Selenium, Multiprocessing, PyTorch — *July 2022 – Jan. 2023*

- Engineered an automated web-crawling system using Selenium and Python multiprocessing to gather and process gigabytes of sales and merchandise data from 50+ sellers of a competitor platform.
- Designed and implemented the backend database (PostgreSQL) to store and query all competitor data, building the foundation for a new revenue estimation system.

OPEN SOURCE CONTRIBUTIONS

Flyte: Open-source ML Workflow Orchestration Platform | Go, Python, Kubernetes, Protobuf — *Dec. 2024 – Present*

- Implemented vertical pod scaling for Flyte to auto-recover from OOM errors by dynamically adjusting memory allocations on task retries, contributing code across Flytekit, FlyteAdmin, FlytePropeller, and FlytePlugins and adding robust unit tests (PR:#6293)
- Developed and integrated registration result collection, output formatting, and quiet mode functionality to Flytekit, enabling registered entity summary option and quiet flag for pyflyte register (PR:#3028)

SELECTED PROJECTS

FAWN: A Chain-Replicated Key-Value Storage System | Rust, gRPC, Protobuf, Tokio — *May 2025 – June 2025*

- Built the core frontend and backend components of a distributed key-value store based on the FAWN architecture, implementing the chain replication protocol and a simplified routing logic to reduce coordination complexity.
- Designed and implemented a novel stage-based execution model for backend nodes to handle dynamic joins safely, ensuring data consistency and replication correctness during reconfiguration.

Chord: A P2P Scalable Distributed File System | AWS, RPC, C++ — *March 2023 – May 2023*

- Implemented a Chord-based distributed file system with auto-scaling capabilities on AWS EC2, utilizing CloudWatch and Auto Scaling Groups to dynamically adjust system capacity based on storage usage.
- Integrated advanced features including file chunking, 3x replication, and automatic data migration, enhancing load balancing and fault tolerance while maintaining high availability.